



# Agathiya Tharun

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🏠 Maryland

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## EDUCATION:

**UCLA** | MS Mech/Aero Engineering | **Class of 2026** | **GTA** for Advanced Topics in MATLAB Programming | **4.00 GPA**  
Graduate Researcher at **RoMeLa** (Robotics & Mechanisms Laboratory) under Dr. Dennis Hong; **Machine Vision** sub team

**Purdue University** | BS Mechanical Engr; Minor in Manufacturing Engr | **Top 12.5% of Seniors** | Class of 2024 | **3.83 GPA**  
6X Semester Honors & Dean's List | **Studied Abroad** in Singapore - Bioinspired Materials and Structures | **2X Startup Cofounder**

## SKILLS:

|                  |            |         |          |         |              |             |         |             |           |
|------------------|------------|---------|----------|---------|--------------|-------------|---------|-------------|-----------|
| <b>General:</b>  | GD&T       | DFM     | CAM      | R&D     | Machine Shop | Prototyping | FEA     | 3D Printing | Mfg.      |
| <b>CAD:</b>      | SolidWorks | NX      | Inventor | CREO    | CATIA V6/5   | Fusion360   |         |             |           |
| <b>Software:</b> | MATLAB     | Python  | C        | Java/JS | SQL          | VBA         | HTML    | CSS         | AngularJS |
| <b>Other:</b>    | LabVIEW    | Arduino | R-PI     | PUTTY   | VeriCut      | AFDEX       | Inspire | OpenSim     | Jira      |

## EXPERIENCES:

**Tesla – Drive Systems Mechanical Design Engineer** | Geartrain Intern MAY 13, 2024 – AUGUST 16, 2024

- Designed a micron-accurate test rig to support Cybertruck drive units spun at 12k RPM to analyze their vibration signatures.
  - Used extensive DFM and GD&T to allow the rig to be modular, multi-purpose, easy to use, and support lubrication.
  - Designed and executed experiments on Cybertruck's differential lockers to catch a faulty system before **a driver loses control**.
- Worked heavily with electro-mechanical systems, diff-lockers, MATLAB, DAQ units, and CAN buses.

**Boeing - Product Development Engineer** | Weights Intern MAY 30, 2023 – AUGUST 18, 2023

- Led cross-functional initiatives to develop a new high-wing aircraft by advising and defending decisions with loadability analysis.
- Optimized CG envelope study efforts in an airplane loadability analysis tool with VBA. Presented to an executive committee
- Optimized a 777-9 propulsion duct bracket with Inspire's Generative Design and FEM Analysis for weight savings of 29.5%.

**SpaceX - Fairing Refurbishment Engineer** | Launch Intern MAY 9, 2022 – AUGUST 5, 2022

- Supported Falcon 9/Heavy campaigns through executing refurbishment operations, creating/dispositioning issue tickets, defending engineering analysis, and designing refurb planning related to pneumatic, mechanical, and control systems.
- Conducted material analysis, high-level design, and deep vendor screening to develop reusable fairing bagging to **save \$764K/yr**.
- Designed, stress analyzed, and mass-produced (**446% savings**) foldable fairing stand extension handles to improve safety & agility.

**NASA Resilient Extra-Terrestrial Habitats Institute (RETHI)** | Research Position JANUARY 19, 2021 – MAY 5, 2023  
Purdue-led project at the Ray Herrick Labs to design/prototype modular vacuum-screwdriver end effectors to allow autonomous robots to repair issues on extraterrestrial habitats. Presented [findings](#) at the Purdue Research Conference. My research was highlighted on all Purdue's social media including the [official website](#), [YouTube](#), [Facebook](#), [Instagram](#), [LinkedIn](#), and [Twitter](#).

**Designed With Purpose - Design Engineer** | Internship MAY 17, 2021 – AUGUST 1, 2021

- Produced static and dynamic luxury furniture through learned machine shop skills and craftsmanship. Video documented [projects](#).
- Learned safe machine operation and maintenance table saw, bandsaw, sanders, pantograph/domino routers, lathe, joiner/planar, etc.

**Lockheed Martin - Software Engineer** | Summer & Co-Op Internship JULY 1, 2019 – AUGUST 1, 2020  
Developed back-end functionality of security forms and approval workflows. Designed website aesthetic and functionality.

## ACTIVITIES:

- Community:** Purdue Admissions **Tour Guide Employee**, BGR **Team Leader** Volunteer, Habitat for Humanity **Volunteer**
- Leadership:** **Eagle Scout**, Cary Hall Unit **President**, NTHS **President and Founder**, Purdue Goss Scholars **Leadership** Council
- Notable Activities:** ASME Podcast **Director and Host**, ASME **Design Lead**, Purdue Space Program

## ACHIEVEMENTS:

**Eagle Scout** | NESA Member ('20)

**Lockheed Martin STEM Scholarship**

**1 of 10 Eli Lilly** Scholarship ('23)

**Top 5** in Ventures Pitch Competition ('24)

**Purdue GRIT Award** ('24)

**Top 11** in Malott Competition ('24)

**IN Space Grant** Researcher Award ('21)

**Top 25 Student** ('20)

Certificate of Meritorious Service (600 hrs)

Tamil Fluency - **Seal of Biliteracy** ('20)

**3 Arduino** Certificates ('21-'23)

**FEA, Python, GD&T** Certificates ('23)